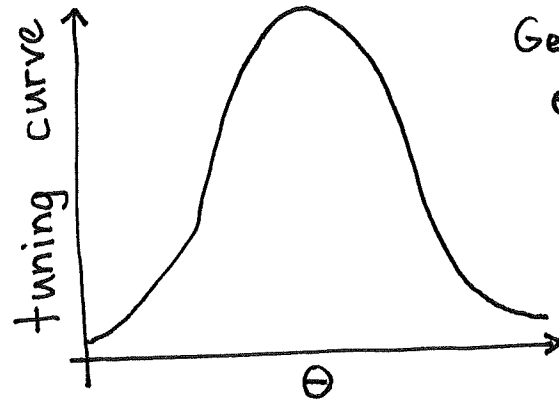
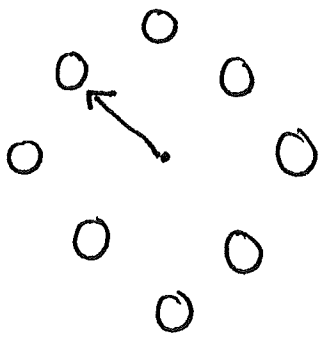


Low-dimensional dynamics in motor cortex:

Neural tuning in center out reach task:

Reaching task



Georgopoulos et. al

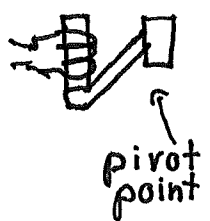
Monkeys repeatedly reach from central location to peripheral locations (typically 8 possibilities). Many neurons are tuned to position (e.g. reach direction), but other neurons do not have clear tuning for ~~many~~ movement parameter.

The more modern approach is to quantify how motor cortex dynamics facilitate the generation of body dynamics.

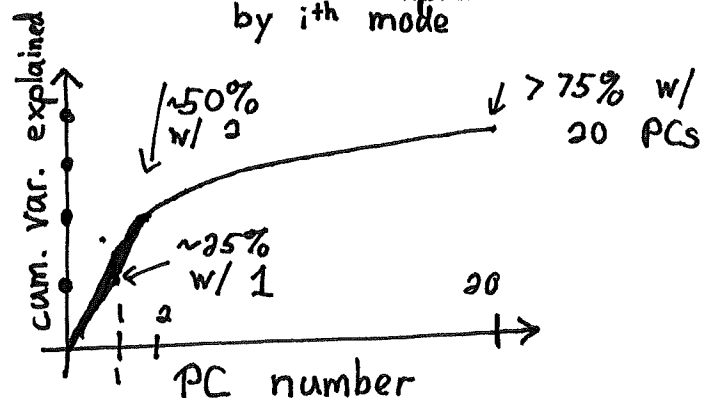
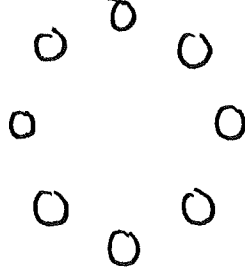
A critical observation is that neural activity is often low-dimensional. Variance decomposition:

$$\mathbf{h} = \sum_{i=1}^N \mathbf{P}_i \xi^{(i)} \Rightarrow \langle \mathbf{h}^T \mathbf{h} \rangle = \sum_{i=1}^N \underbrace{\langle \mathbf{P}_i^2 \rangle}_{\text{Variance explained by } i^{\text{th}} \text{ mode}} = \sum_{i=1}^N \lambda_i$$

Wrist task



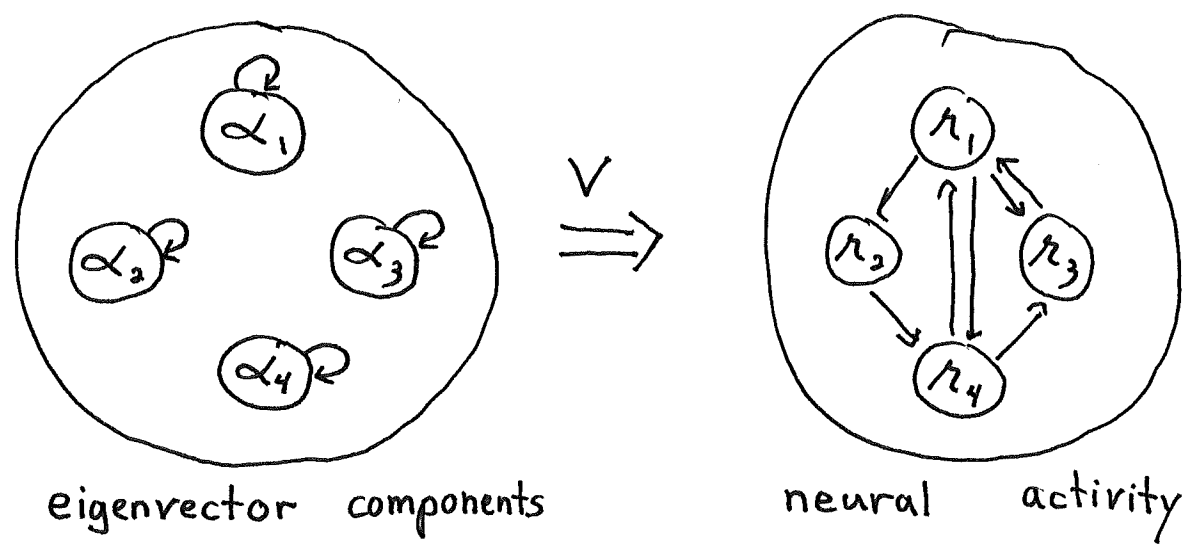
Targets



Similar PCA based analyses are very widespread in systems neuroscience.

PCA-like dimensionality reduction can become more motor cortex like by explicitly adding dynamics.

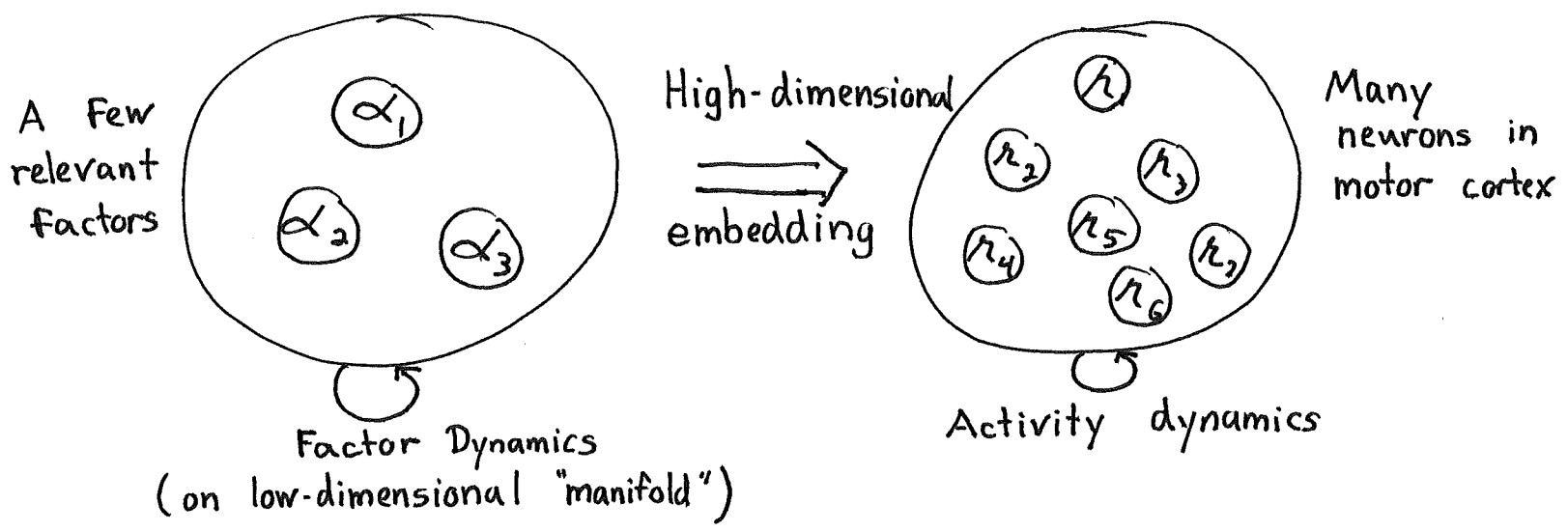
Linear neural network perspective from previous lecture



More general perspective:

Latent Factor Space

Neural Activity Space



Gaussian Process Factor Analysis (GPFA)

Probabilistic model linking latent space to activity

$$P(r_t | \alpha_t) = \mathcal{N} \left(\underbrace{C}_{\text{Coupling of neuron to mode}} \underbrace{\alpha_t}_{\text{Single timepoint, all neurons \& factors}} + \underbrace{d}_{\text{offset of neuron}}, \underbrace{\Sigma}_{\text{Noise level of neuron}} \right)$$

Generative model for smooth neural trajectories:

$$P(\alpha_i) = \mathcal{N}(0, K_i)$$

Single factor, all time points Temporal covariance matrix.

$$K_{i,t_1,t_2} = \underbrace{\sigma_{n,i}^2}_{\text{White noise}} \delta_{t_1,t_2} + \underbrace{\sigma_{s,i}^2}_{\text{Signal strength}} e^{-\underbrace{(t_1-t_2)^2/(2\tau_i^2)}_{\text{controls correlation time}}}$$

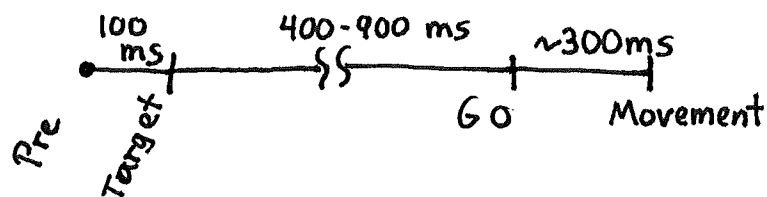
Overall scale of α is arbitrary due to C. Set,

$$\sigma_{n,i}^2 + \sigma_{s,i}^2 = 1$$

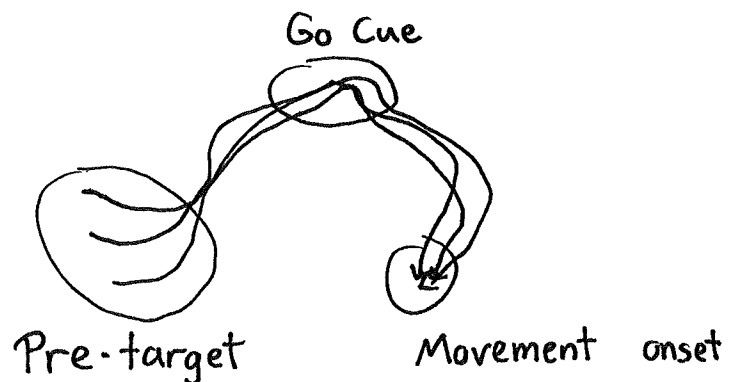
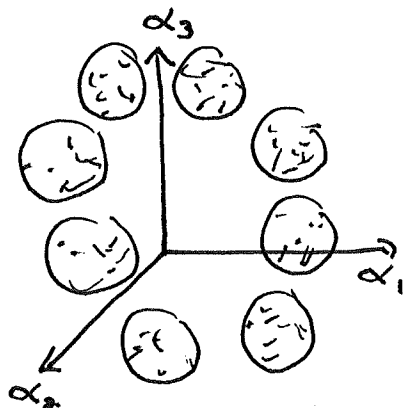
Fit model parameters with maximum likelihood

$$\ell(c, d, \Sigma, \sigma_n^2, \tau) = \log P(r | c, d, \Sigma, \sigma_n^2, \tau)$$

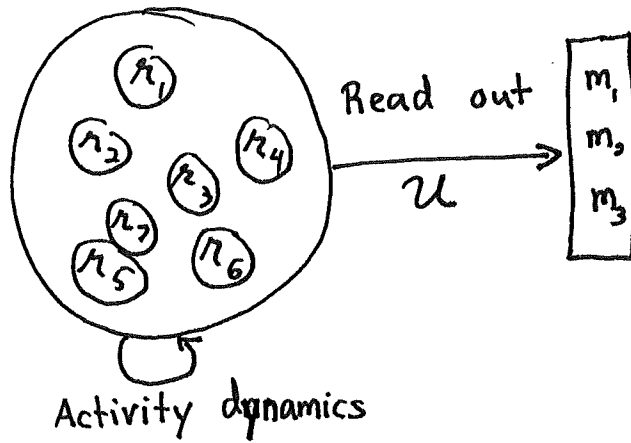
Applied to motor cortex during delayed reach task



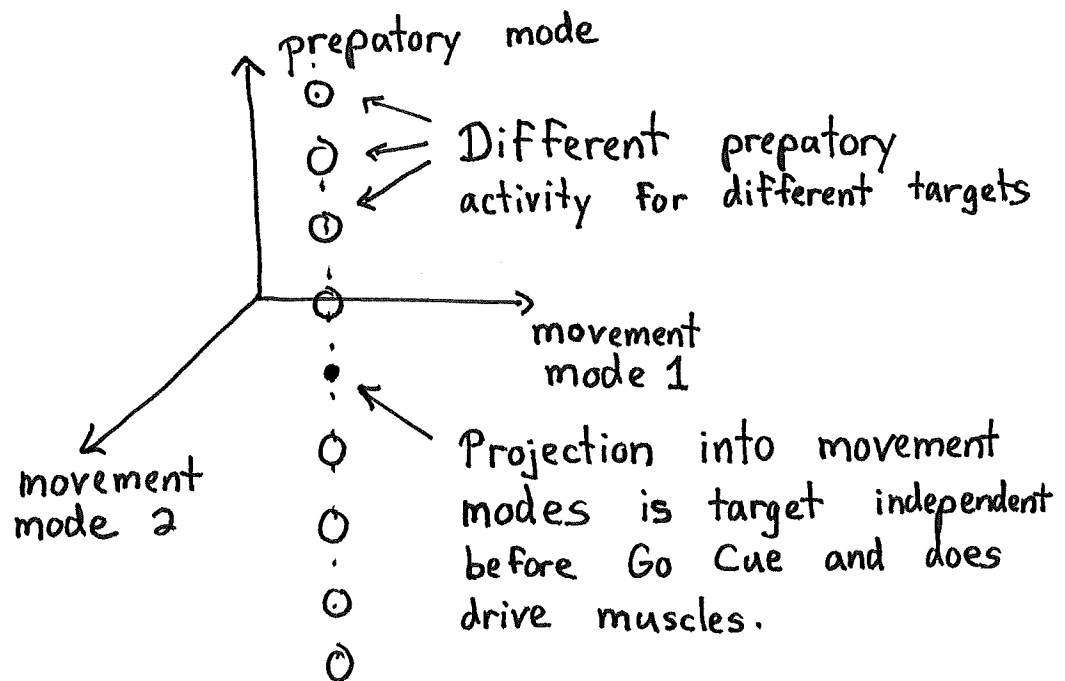
Factor analysis (i.e., GPFA with $\sigma_s^2 = 0$) reveals a low-dimensional space of preparatory activity, and GPFA constructs coherent trajectories throughout task.



Neural dynamics after target onset and before go cue lead to a tightening of neural trajectories and less trial-to-trial. Accordingly, movements are faster & more accurate. Preparatory activity can be conceptualized as "null space" activity that is not "read out" by muscles.



$$m = ur, \quad m(t) = 0 \Rightarrow (ur)(t) = 0 \not\Rightarrow r(t) = 0$$



Simplest linear read out case.

$$[u] = N \times M, \quad M < N, \quad (N-M)\text{-dimensional null space with}$$

$$ur = 0$$